ML PROJECT REPORT

Integrated EV Battery Intelligence System

NASA Battery Dataset — B0005 / B0006 / B0007 / B0018

Anomaly Detection → SoC Estimation → SoH Prediction → RUL Prediction → EV Range Prediction

1. Executive Summary

This report documents the architecture, design decisions, and methodology of an integrated machine learning system for EV battery intelligence, built on the NASA battery degradation dataset. The system estimates four critical battery health indicators — State of Charge (SoC), State of Health (SoH), Remaining Useful Life (RUL), and predicted driving range in kilometres — using a multi-stage deep learning pipeline.

The pipeline applies an Isolation Forest for anomaly removal, engineers domain-specific features from raw discharge cycle data, and trains three independent deep learning models. SoC and SoH use a CNN–BiLSTM–Attention architecture; RUL uses a Transformer–LSTM hybrid; and a DNN + GBM ensemble predicts the final driving range.

2. Dataset

The NASA Prognostics Center of Excellence Battery dataset contains four lithium-ion batteries (B0005, B0006, B0007, B0018), cycled at room temperature until capacity fell below 1.4 Ah (end-of-life). Only discharge cycles are used. Four measurements are extracted per time-step:

- Voltage measured (V) — terminal voltage
- Current measured (A) — discharge current
- Temperature measured (°C) — cell surface temperature
- Capacity (Ah) — measured discharge capacity per cycle

3. Pipeline Overview

The system is structured as a five-stage sequential pipeline. Stages 4a, 4b, and 4c run in parallel after shared preprocessing.

Step	Stage	Method	Output
1	Data loading	scipy.io.loadmat, discharge filter	Raw DataFrame (4 batteries concatenated)
2	Anomaly detection	Isolation Forest (n=200, contam=0.03)	clean_df — ~3% anomalous rows removed
3	Feature engineering	SoC / SoH / RUL labels + derived signals	Labelled DataFrame with power, diffs
4a	SoC estimation	CNN–BiLSTM–Attention	SoC_predicted ∈ [0, 1]
4b	SoH prediction	CNN–BiLSTM–Attention	SoH_predicted ∈ [0, 1]
4c	RUL prediction	Transformer encoder + LSTM	RUL_predicted (cycles remaining)
5	Range prediction	DNN + GBM ensemble (50:50 blend)	range_km — actionable km output

4. Preprocessing

4.1 Anomaly Detection — Isolation Forest

Before modelling, the raw DataFrame is screened with Isolation Forest (scikit-learn). Records assigned a score of −1 are discarded, removing approximately 3% of discharge timesteps exhibiting implausible voltage or capacity readings.

- n_estimators = 200 — stable scoring ensemble
- contamination = 0.03 — matches expected sensor fault rate

- random_state = 42 — reproducibility

4.2 Feature Engineering

Three target labels are computed before training:

```
SoC = Capacity_at_timestep / max(Capacity_in_cycle)     [clipped to 0,1]
SoH = mean_capacity_of_cycle / initial_mean_capacity    [clipped to 0,1]
RUL = EOL_cycle - current_cycle                         [clipped at 0]
```

Additional derived signals: power = V × |I|, voltage_x_current = V × I, temp_diff = Δtemperature, voltage_diff = Δvoltage.

4.3 Sequence Construction

All features are scaled to [0, 1] with MinMaxScaler. A sliding window of length 30 produces tensors of shape (N, 30, n_features). Separate scalers are used for features (scaler_X) and targets (scaler_y) to allow independent inverse transformation at inference.

5. Model Architectures

5.1 SoC Estimation — CNN–BiLSTM–Attention

- Conv1D × 3 (128 filters, kernel=3, ReLU, BatchNorm) with residual skip after 3rd layer
- BiLSTM(128, return_sequences=True) → BiLSTM(64, return_sequences=True)
- Attention: Dense(1, tanh) → Softmax → Multiply → reduce_sum over time
- Head: Dense(128, ReLU) + Dropout(0.3) → Dense(64, ReLU) + Dropout(0.2) → Dense(1, linear)
- Loss: Huber | Optimizer: Adam lr=3e-4

5.2 SoH Prediction — CNN–BiLSTM–Attention

Same topology as SoC but independently trained with separate weights:

- Conv1D(64) + Conv1D(128) — no residual connection in SoH variant
- BiLSTM(128) → BiLSTM(64) → same Attention mechanism
- Head: Dense(64) → Dense(32) → Dense(1, sigmoid) — bounded [0,1] output
- Loss: MSE | Optimizer: Adam lr=1e-3

5.3 RUL Prediction — Transformer–LSTM Hybrid

- Input projection: Dense(64) maps n_features to d_model=64
- 2× Transformer encoder blocks: MultiHeadAttention(heads=4, key_dim=16) + Add & LayerNorm + FFN(128) + Add & LayerNorm
- LSTM(64, return_sequences=False) — summarises degradation trajectory from Transformer output
- Head: Dense(32, ReLU) + Dropout(0.1) → Dense(1, linear)
- Loss: MSE | Optimizer: Adam

5.4 EV Range Prediction — DNN + GBM Ensemble

Inputs: SoC, SoH, and RUL predictions from upstream models plus original electrical measurements.

```
range_km = base_range × SoC × SoH × temp_factor   (base_range = 300 km)
temp_factor = 1.0 - 0.003 × |temperature - 25|
```

- DNN: Dense(128) + 2 residual blocks (64→64, 32→32 with BatchNorm) + Dense(1, linear)
- GBM: sklearn GradientBoostingRegressor — captures non-linear tabular interactions
- Blend: range_km = scaler_y.inverse_transform(0.5 × DNN_pred + 0.5 × GBM_pred)

6. Training Configuration

Parameter	SoC model	SoH model	RUL model
Sequence length	30	30	30
Batch size	64	64	64
Max epochs	100	100	100

Optimizer	Adam (lr=3e-4)	Adam (lr=1e-3)	Adam (default)
Loss function	Huber	MSE	MSE
Early stopping	patience=15	patience=15	patience=15
LR reduction	factor=0.5, pat=7	factor=0.5, pat=7	factor=0.5, pat=7
Train / val split	80 / 20	80 / 20	80 / 20

All models share the same callbacks: EarlyStopping monitors val_loss with patience=15 and restores best weights; ReduceLROnPlateau halves the learning rate when val_loss stagnates for 7 epochs, with a floor of lr=1e-6.

7. Evaluation & Inference

7.1 Metrics

- MAE — average absolute deviation in original physical units
- RMSE — penalises large errors more heavily; sensitive to outlier cycles
- R^2 — proportion of variance explained; 1.0 = perfect fit

Each model produces a time-series overlay (first 500 validation samples) and a parity scatter plot. Both are saved as .png during pipeline execution.

7.2 Single-Point Inference

The predict_range_single() function accepts instantaneous voltage, current, temperature, SoC, SoH, and RUL values, applies scaler_X, runs both DNN and GBM, blends 50:50, inverse-transforms with scaler_y, and clips to ≥ 0 km. Designed for real-time EV dashboard integration where a new battery reading arrives every few seconds.

8. Key Design Decisions

Decision	Choice	Rationale
SoC & SoH architecture	CNN–BiLSTM–Attention	CNN extracts local patterns; BiLSTM captures bidirectional temporal dependencies; Attention weights informative timesteps
RUL architecture	Transformer–LSTM	Transformer captures long-range inter-cycle relationships via global self-attention; LSTM models degradation trajectory
Range prediction	DNN + GBM 50:50	DNN learns smooth non-linear mappings; GBM captures tabular feature interactions; blending reduces variance
SoC loss function	Huber loss	Robust to occasional noisy discharge cycles unlike MSE, which over-penalises outlier readings
SoH output activation	Sigmoid	Enforces output to [0,1], consistent with SoH being a bounded normalised degradation metric
Anomaly detection	Isolation Forest	Unsupervised, dataset-agnostic; no labelled anomalies needed; contamination=0.03 matches expected NASA sensor fault rate

9. Limitations & Future Work

- Range label is a physics-inspired proxy — no actual odometer data in NASA dataset; vehicle telemetry should replace the heuristic in production.
- Transformer and BiLSTM models are sensitive to seq_len=30; longer sequences may improve RUL at higher computational cost.
- Models trained on four batteries from one experimental batch; generalisation to different chemistry (NMC vs LFP) requires transfer learning or retraining.

- The 50:50 DNN–GBM blend weight is fixed; stacking to learn the weight could improve range accuracy.
- Temperature effect uses a linear factor; Arrhenius-based modelling would improve cold-weather range estimates.

10. Conclusion

This project delivers a complete, production-oriented ML pipeline for EV battery intelligence. Isolation Forest anomaly filtering feeds three independently trained deep learning models — CNN–BiLSTM–Attention for SoC and SoH, Transformer–LSTM for RUL — whose outputs are aggregated by a DNN + GBM ensemble into a directly actionable range prediction.

The architecture is modular by design: each stage produces standardised predictions that can be replaced or retrained independently. Early stopping, learning rate reduction, and residual connections ensure training stability on the relatively small NASA dataset, while ensemble blending guards against single-model bias.